GRADE 11 · ALGEBRA II · MID-TERM · STUDY GUIDE

Mid-Term Study Guide: Chapters 5–6

Radical, Exponential and Logarithmic Functions
Lessons 5-2 to 6-5: read, copy, practise, then sit the mock

How to use this guide

The exam: 55 minutes, 20 marks. Section A: 15 multiple-choice questions (0.5 mark each). Section B: 4 matching pairs. Section C: 4 fill-in-the-blank items. Then short-answer questions and one problem-solving question with a table and a graph sketch.

Coverage: Chapter 5 Rational Exponents and Radical Functions (Lessons 5-2 to 5-6) and Chapter 6 Exponential and Logarithmic Functions (Lessons 6-1, 6-3, 6-4 and 6-5).

Plan: (1) read one lesson and copy each example without looking; (2) do the practice for that chapter; (3) sit the mock exam in 55 minutes with no notes; (4) mark it with the key and redo the examples for every lost mark.

Chapter 5: Rational Exponents and Radical Functions

Lesson 5-2: Properties of exponents and radicals

$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$ (the *denominator* is the root). $a^{-n} = \dfrac{1}{a^n}$. $a^m a^n = a^{m+n}$, $\dfrac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$, $(ab)^n = a^n b^n$, $\sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$.

Example 1: rational exponents

$27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$. $16^{3/4} = \left(\sqrt[4]{16}\right)^3 = 2^3 = 8$. $8^{-1/3} = \dfrac{1}{8^{1/3}} = \dfrac{1}{2}$.

Trap: take the root first, then the power. $27^{2/3}$ is not $27 \cdot \frac{2}{3}$.

Example 2: simplify with exponent rules

$(8x^6)^{1/3} = 8^{1/3}\,x^{6/3} = 2x^2$. $x^{1/2} \cdot x^{3/2} = x^2$. $\dfrac{x^{5/3}}{x^{2/3}} = x^1 = x$.

Example 3: simplify radicals

$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$. $\sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$. $\sqrt[4]{x^3} = x^{3/4}$. Rationalise: $\dfrac{6}{\sqrt{3}} = \dfrac{6\sqrt{3}}{3} = 2\sqrt{3}$.

Lesson 5-3: Graphing radical functions

Parent $y = \sqrt{x}$: domain $x \geq 0$, range $y \geq 0$, points $(0,0), (1,1), (4,2), (9,3)$. For $f(x) = a\sqrt{x-h} + k$: the graph starts at (h,k); domain $x \geq h$; range $y \geq k$ if $a > 0$ and $y \leq k$ if $a < 0$. The cube-root function $y = \sqrt[3]{x}$ has domain and range all real numbers.

Example 4: $g(x) = \sqrt{x+4} - 3$

Left 4, down 3: start $(-4, -3)$; domain $x \geq -4$; range $y \geq -3$. y-intercept: $g(0) = \sqrt{4} - 3 = -1$. x-intercept: $\sqrt{x+4} = 3$ gives $x = 5$.

Example 5: $g(x) = -\sqrt{x-1} + 2$

Reflected in the x-axis, right 1, up 2: start $(1, 2)$; domain $x \geq 1$; range $y \leq 2$. x-intercept: $\sqrt{x-1} = 2$ gives $x = 5$.

Trap: $x + 4$ moves the graph *left*, and the domain starts at -4, not 4.

Lesson 5-4: Solving radical equations

(1) Isolate the radical. (2) Raise both sides to the power equal to the index. (3) Solve. (4) **Check every answer in the original equation.** Squaring can create an **extraneous solution**.

Example 6: $\sqrt{x+3}=5$

$x+3=25$, so $x=22$. Check: $\sqrt{25}=5$. ✓

Example 7: $\sqrt{x+7}=x+1$

Square: $x+7=x^2+2x+1$, so $x^2+x-6=0$, $(x+3)(x-2)=0$. Check $x=2$: $\sqrt{9}=3=2+1$. ✓ Check $x=-3$: $\sqrt{4}=2$ but $-3+1=-2$. Extraneous. Only $x=2$.

Trap: $(x+1)^2=x^2+2x+1$, not x^2+1.

Example 8: other types

$\sqrt[3]{x-2}=3$: cube both sides, $x-2=27$, so $x=29$. $\sqrt{3x-2}=\sqrt{x+6}$: $3x-2=x+6$, so $x=4$; check $\sqrt{10}=\sqrt{10}$. ✓

Lesson 5-5: Function operations and composition

$(f+g)(x)=f(x)+g(x)$, $(f-g)(x)$, $(fg)(x)=f(x)\cdot g(x)$, $\left(\frac{f}{g}\right)(x)=\frac{f(x)}{g(x)}$ with $g(x)\neq 0$. **Composition:** $(f\circ g)(x)=f(g(x))$: the inside function acts first. In general $f\circ g\neq g\circ f$.

Example 9: $f(x)=3x-1$ and $g(x)=x^2$

$(f+g)(x)=x^2+3x-1$. $(fg)(x)=3x^3-x^2$. $(f\circ g)(x)=f(x^2)=3x^2-1$. $(g\circ f)(x)=(3x-1)^2=9x^2-6x+1$.

Example 10: values and domains

$f(x)=x+1$, $g(x)=2x$: $f(g(3))=f(6)=7$ but $g(f(3))=g(4)=8$.

$f(x)=x+4$, $g(x)=x-2$: $\left(\frac{f}{g}\right)(x)$ has domain $x\neq 2$.

$f(x)=\sqrt{x}$, $g(x)=x-5$: $(f\circ g)(x)=\sqrt{x-5}$ has domain $x\geq 5$.

Lesson 5-6: Inverse functions

f^{-1} undoes f: $f^{-1}(f(x))=x$ and $f(f^{-1}(x))=x$. **To find it:** write $y=f(x)$, swap x and y, solve for y. The graphs of f and f^{-1} are reflections in $y=x$, so (a,b) on f means (b,a) on f^{-1}. f has an inverse function only if it is **one-to-one** (passes the horizontal line test).

Example 11: $f(x)=\frac{2x+1}{5}$

Swap: $x=\frac{2y+1}{5}$. Then $5x=2y+1$, so $y=\frac{5x-1}{2}$ and $f^{-1}(x)=\frac{5x-1}{2}$. Check: $f\left(\frac{5x-1}{2}\right)=\frac{(5x-1)+1}{5}=x$.

Trap: $f^{-1}(x)$ is not $\frac{1}{f(x)}$.

Example 12: restricted domain and a table

$f(x)=x^2$ is not one-to-one, but with $x\geq 0$ its inverse is $f^{-1}(x)=\sqrt{x}$. For $f(x)=\sqrt{x+3}$: $x=\sqrt{y+3}$ gives $y=x^2-3$ with $x\geq 0$.

If $f(1)=4$, $f(2)=7$, $f(3)=10$, then $f^{-1}(7)=2$. Cube-type: $f(x)=x^3+1$ gives $f^{-1}(x)=\sqrt[3]{x-1}$.

Chapter 6: Exponential and Logarithmic Functions

Lesson 6-1: Key features of exponential functions

$f(x) = a \cdot b^x$ with $a \neq 0$, $b > 0$, $b \neq 1$. Growth if $b > 1$; decay if $0 < b < 1$. y-intercept $(0, a)$. Domain: all reals. Horizontal asymptote $y = 0$; for $f(x) = a\,b^x + k$ it is $y = k$. Percent change: growth factor $b = 1 + r$, decay factor $b = 1 - r$.

Example 13: features

$f(x) = 5 \cdot 3^x$: growth, $f(0) = 5$, $f(2) = 45$. $f(x) = 2^x + 1$: asymptote $y = 1$, y-intercept $f(0) = 2$, range $y > 1$.

Example 14: models

A population of 8000 grows 5% a year: $P(t) = 8000(1.05)^t$, so $P(3) = 8000(1.157625) = 9261$.
A car worth $60\,000$ SAR loses 12% a year: $V(t) = 60000(0.88)^t$, so $V(2) = 60000(0.7744) = 46\,464$ SAR.

Example 15: write the function from two points

Through $(0, 3)$ and $(2, 12)$: $a = 3$, and $3b^2 = 12$ gives $b^2 = 4$, so $b = 2$. $f(x) = 3 \cdot 2^x$.

Trap: the growth factor for 5% is 1.05, not 0.05.

Lesson 6-3: Logarithms

$\log_b x = y$ means $b^y = x$ $(b > 0,\ b \neq 1,\ x > 0)$. Special values: $\log_b 1 = 0$ and $\log_b b = 1$. **Common logarithm:** $\log x = \log_{10} x$. **Natural logarithm:** $\ln x = \log_e x$.

Example 16: evaluate and convert

$\log_4 64 = 3$ (since $4^3 = 64$). $\log_2 \dfrac{1}{8} = -3$. $\log_9 3 = \dfrac{1}{2}$. $\log 1000 = 3$. $\ln e^3 = 3$. $2^5 = 32 \iff \log_2 32 = 5$.

Example 17: solve

$\log_2 x = 5$ gives $x = 2^5 = 32$. $\log_x 49 = 2$ gives $x = 7$.
Exponential model: $2^x = 50$ gives $x = \log_2 50 = \dfrac{\ln 50}{\ln 2} \approx 5.644$.

Lesson 6-4: Logarithmic functions

$f(x) = \log_b x$ is the inverse of b^x. Domain $x > 0$; range all reals; passes through $(1, 0)$ and $(b, 1)$; vertical asymptote $x = 0$; increasing if $b > 1$. For $g(x) = \log_b(x - h) + k$: domain $x > h$, asymptote $x = h$.

Example 18: $g(x) = \log_2(x + 1) - 3$

Left 1, down 3. Domain $x > -1$; asymptote $x = -1$. x-intercept: $\log_2(x + 1) = 3$ gives $x + 1 = 8$, so $x = 7$.

Example 19: inverses and rate of change

$f(x) = 3^x + 2$: $x = 3^y + 2$, so $f^{-1}(x) = \log_3(x - 2)$. $g(x) = \log_5 x - 1$: $g^{-1}(x) = 5^{x+1}$.
Average rate of change of $\log_2 x$ from $x = 2$ to $x = 8$: $\dfrac{3 - 1}{8 - 2} = \dfrac{1}{3}$.

Lesson 6-5: Properties of logarithms

$\log_b(MN) = \log_b M + \log_b N$. $\log_b \dfrac{M}{N} = \log_b M - \log_b N$. $\log_b M^p = p \log_b M$. **Change of base:** $\log_b x = \dfrac{\log x}{\log b} = \dfrac{\ln x}{\ln b}$.

Example 20: evaluate

$\log_2 8 + \log_2 4 = 3 + 2 = 5 = \log_2 32$. $\log_3 54 - \log_3 2 = \log_3 27 = 3$. $3 \log_2 4 = \log_2 64 = 6$.

Example 21: expand, condense, change base

$\log \dfrac{x^2 y}{z^3} = 2\log x + \log y - 3\log z$. $\quad 2\log_3 x + \log_3 5 = \log_3 (5x^2)$. $\quad \log_4 32 = \dfrac{\log_2 32}{\log_2 4} = \dfrac{5}{2}$.

Trap: $\log(M+N) \neq \log M + \log N$ and $\dfrac{\log M}{\log N} \neq \log \dfrac{M}{N}$.

Practice (answers at the end)

Chapter 5

1. Evaluate $64^{2/3}$ and $125^{-1/3}$.

2. Simplify $(81x^8)^{3/4}$ and $\sqrt{98}$.

3. State the start point, domain and range of $f(x) = \sqrt{x-3} + 2$.

4. Solve $\sqrt{4x+1} = x + 1$ and check for extraneous solutions.

5. Solve $\sqrt[3]{2x+3} = -3$.

6. With $f(x) = x - 2$ and $g(x) = x^2 + 1$ find $(f \circ g)(x)$ and $(g \circ f)(x)$.

7. State the domain of $\dfrac{f}{g}$ for $f(x) = x^2$ and $g(x) = x - 6$.

8. Find the inverse of $f(x) = 3x + 7$ and of $g(x) = \dfrac{x-2}{4}$.

Chapter 6

9. A bike loses 15% of its value each year and costs 2000 SAR new. Write $V(t)$ and find $V(2)$.

10. Write $\log_5 125 = 3$ in exponential form; evaluate $\log_3 \dfrac{1}{27}$.

11. Find the domain and asymptote of $g(x) = \log_3(x-4) + 1$.

12. Find the inverse of $f(x) = 2^x - 5$.

13. Evaluate $\log_6 4 + \log_6 9$.

14. Write $\log_2 \dfrac{8x}{y^2}$ as a sum and difference of logs.

Answers to practice

Ch 5. 1. 16; $\dfrac{1}{5}$. 2. $27x^6$; $7\sqrt{2}$. 3. Starts $(3, 2)$; domain $x \geq 3$; range $y \geq 2$. 4. $4x+1 = x^2+2x+1$, so $x(x-2)=0$; both $x=0$ and $x=2$ check ($1=1$, $3=3$). 5. $2x+3=-27$, $x=-15$. 6. $(f\circ g)(x) = x^2 - 1$; $(g\circ f)(x) = (x-2)^2+1 = x^2-4x+5$. 7. $x \neq 6$. 8. $f^{-1}(x) = \dfrac{x-7}{3}$; $g^{-1}(x) = 4x + 2$.

Ch 6. 9. $V(t) = 2000(0.85)^t$; $V(2) = 2000(0.7225) = 1445$ SAR. 10. $5^3 = 125$; -3. 11. Domain $x > 4$; asymptote $x = 4$. 12. $f^{-1}(x) = \log_2(x+5)$. 13. $\log_6 36 = 2$. 14. $3 + \log_2 x - 2\log_2 y$.

GRADE 11 · ALGEBRA II · MID-TERM · MOCK EXAM · 20 MARKS · 55 MINUTES

Mid-Term Mock Exam: Chapters 5–6

Radical, Exponential and Logarithmic Functions
No notes. Calculator not needed.

Name ..... Class _____ Date _____

Section A: Multiple choice

15 questions, 0.5 mark each

Circle ONE letter per question.

1 What is the value of $32^{3/5}$?

A. 2 **B.** 6

C. 8 **D.** $\dfrac{96}{5}$

2 The table shows values of an exponential function f. Which rule fits the table?

x	0	1	2	3
$f(x)$	54	18	6	2

A. $f(x) = 54\left(\dfrac{1}{3}\right)^x$ **B.** $f(x) = 54(3)^x$

C. $f(x) = 18\left(\dfrac{1}{3}\right)^x$ **D.** $f(x) = 54 - 36x$

3 What is the domain of $f(x) = \sqrt{x+6} - 2$?

A. $x \geq 6$ **B.** $x \geq -2$

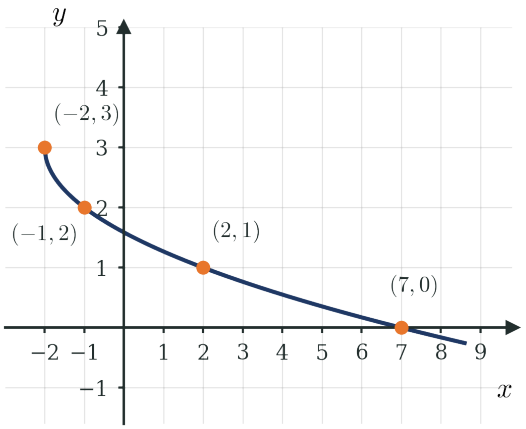
C. $x \geq 2$ **D.** $x \geq -6$

4 For all $x > 0$ and $y > 0$, which statement is true?

A. $\log(x+y) = \log x + \log y$ **B.** $\log\left(x^3 y\right) = 3\log x + \log y$

C. $\log \dfrac{x}{y} = \dfrac{\log x}{\log y}$ **D.** $(\log x)^2 = 2\log x$

5 The graph of f is shown. Which equation defines f?



A. $f(x) = -\sqrt{x+2} + 3$ **B.** $f(x) = -\sqrt{x-2} + 3$

C. $f(x) = \sqrt{x+2} + 3$ **D.** $f(x) = -\sqrt{x+2} - 3$

6 What is the solution of $2\sqrt[3]{x+1} - 3 = 5$?

A. $x = 3$ **B.** $x = 15$

C. $x = 0$ **D.** $x = 63$

7 A solar-farm inverter costing $120\,000$ SAR loses 6% of its value each year. Which expression gives its value after 18 months?

A. $120\,000(0.94)^{1.5}$ **B.** $120\,000(0.94)^{18}$

C. $120\,000(0.06)^{1.5}$ **D.** $120\,000(1.06)^{1.5}$

8 For which pair of functions is $g = f^{-1}$?

A. $f(x) = 3x - 6$ and $g(x) = \dfrac{x}{3} + 6$ **B.** $f(x) = \dfrac{x}{2} + 4$ and $g(x) = 2x - 4$

C. $f(x) = 5x + 10$ and $g(x) = \dfrac{x - 10}{5}$ **D.** $f(x) = 4 - 2x$ and $g(x) = \dfrac{x - 4}{2}$

9 What is the value of $\log_8 32$?

A. $\dfrac{3}{5}$ **B.** $\dfrac{5}{3}$

C. 4 **D.** 2

10 Without solving, which equation has NO real solution?

A. $\sqrt[3]{x+4} = -3$ **B.** $\sqrt{x-4} = 3$

C. $\sqrt{x+4} = -3$ **D.** $\sqrt{2x+1} = 0$

11 What are the domain and vertical asymptote of $g(x) = \log_3(5 - x) + 2$?

A. domain $x > 5$; asymptote $x = 5$ **B.** domain $x > -5$; asymptote $x = -5$

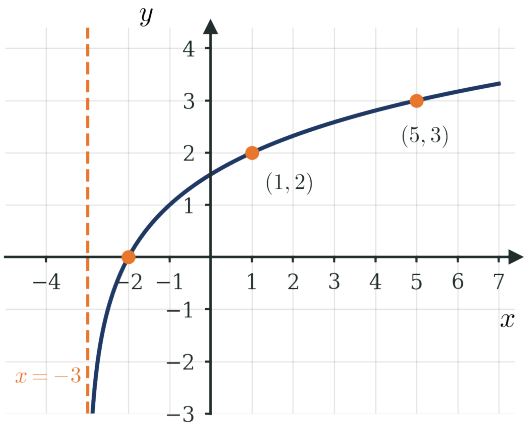
C. domain $x < 5$; asymptote $x = 2$ **D.** domain $x < 5$; asymptote $x = 5$

12 What are the y-intercept and horizontal asymptote of $f(x) = -2(3)^{x-1} + 5$?

A. $(0, -1)$ and $y = 5$ **B.** $\left(0, \dfrac{13}{3}\right)$ and $y = 5$

C. $\left(0, \dfrac{13}{3}\right)$ and $y = -2$ **D.** $(0, 3)$ and $y = 5$

13 The graph of g has a vertical asymptote and passes through the marked points. Which equation defines g?



A. $g(x) = \log_2(x - 3)$ **B.** $g(x) = \log_2 x + 3$

C. $g(x) = \log_2(x + 3)$ **D.** $g(x) = \log_4(x + 3)$

14 What is the inverse of $f(x) = \sqrt[3]{x} + 4$?

A. $f^{-1}(x) = (x - 4)^3$ **B.** $f^{-1}(x) = (x + 4)^3$

C. $f^{-1}(x) = x^3 - 4$ **D.** $f^{-1}(x) = x^3 + 4$

15 Let $f(x) = \sqrt{x+3}$ and $g(x) = x^2 - 7$. What is the domain of $(f \circ g)(x)$?

A. $x \geq 2$ **B.** $-2 \leq x \leq 2$

C. $x \geq -3$ **D.** $x \leq -2$ or $x \geq 2$

Section B: Matching

4 pairs, 0.5 mark each

Match each expression with its value. Write the letter on the line. One value is not used.

a. $\log_2 40 - \log_2 5$ _____

b. $\log_9 27$ _____

c. $2\log_5 10 - \log_5 4$ _____

d. $\log_5 \dfrac{1}{125}$ _____

A. -3

B. $\dfrac{3}{2}$

C. 2

D. 3

E. 5

Section C: Fill in the blank

4 blanks, 0.5 mark each

Complete each statement.

a. The inverse of $f(x) = 5^x$ is $f^{-1}(x) =$ _____.

b. The range of $f(x) = -2\sqrt{x+1} + 3$ is $y \leq$ _____.

c. If $f(a) = b$, then $f^{-1}($_____$) = a$.

d. $\sqrt[5]{x^3} = x$_____ when written with a rational exponent.

Section D: Short answer

Show your working

1 **Error analysis.** A student simplifies $\left(16x^{12}y^{-4}\right)^{3/4}$ and writes $12x^9y^{-3}$.
(a) Find the student's mistake. (b) Write the correct expression with positive exponents only. **[1 mark]**

2 (a) Solve $\sqrt{x+5} + 1 = x$. Isolate the radical first, then square both sides. Find the possible values of x. **[1 mark]**

3 (b) Check each value in the original equation. State which is a solution and which is extraneous. **[1 mark]**

4 Let $f(x) = \sqrt{x+1}$ and $g(x) = x^2 - 1$.
(a) Find $(g \circ f)(x)$ and state its domain.
(b) Find $(f \circ g)(x)$ for all real x. Use it to explain why g is the inverse of f only when g is restricted to $x \geq 0$. **[1.5 marks]**

Section E: Problem solving

4 marks

A dallah of Saudi coffee is poured into a cup at a majlis in Jeddah. The room stays at $30°$C. The coffee cools toward room temperature, and the *excess* above $30°$C halves at a steady rate. Its temperature t minutes after pouring is modelled by

$$T(t) = 30 + A\left(\frac{1}{2}\right)^{t/h},$$

where A and h are constants. The coffee is measured at $T(0) = 94$ and $T(5) = 62$ (degrees Celsius).

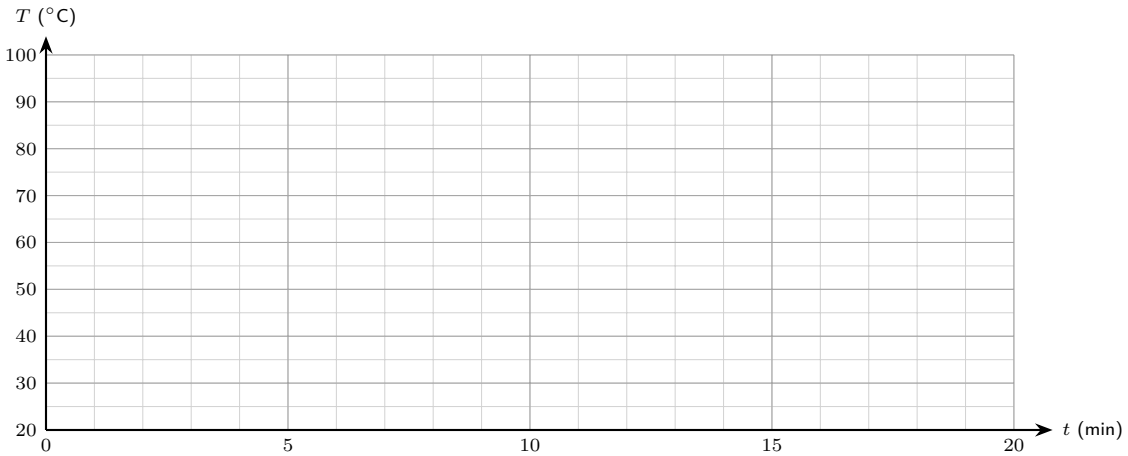
(a) State the horizontal asymptote of T and say what it means for the coffee. **[0.5 mark]**

(b) Use $T(0) = 94$ and $T(5) = 62$ to find A and h. **[1 mark]**

(c) Use the model to complete the table. **[1 mark]**

t (min)	0	5	10	15	20
$T(t)$ (°C)	94	62			

(d) Sketch the graph of T for $0 \le t \le 20$ on the grid. Draw the asymptote as a dashed line, plot the points, and join them with a smooth curve. **[1 mark]**



(e) Hana says: "The coffee lost $32°$C in the first 5 minutes, so after 10 minutes it will be at $30°$C." Explain her mistake. **[0.5 mark]**

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Answer key and marking

Check your work
Mark each part, then revisit the matching example

Section A: multiple choice (0.5 mark each, 7.5 marks)

1 C **2** A **3** D **4** B **5** A **6** D **7** A **8** C **9** B **10** C **11** D **12** B **13** C **14** A **15** D

1. $32^{3/5} = \left(\sqrt[5]{32}\right)^3 = 2^3 = 8$. Errors: 2 drops the power; 6 multiplies the root by 3; $\frac{96}{5}$ multiplies $32 \cdot \frac{3}{5}$.

2. $f(0) = 54$ and each step multiplies by $\frac{1}{3}$. Errors: growth factor for decay; a read from $f(1)$; linear rule.

3. $x + 6 \geq 0$. Errors: sign of h lost ($x \geq 6$); k used as the start ($x \geq -2$); both slips ($x \geq 2$).

4. Product and power rules. Errors: $\log(x + y)$ split; quotient turned into a ratio of logs; $(\log x)^2$ confused with $2 \log x$.

5. Start $(-2, 3)$ gives $h = -2$, $k = 3$; the curve falls, so $a < 0$; check $(2, 1)$: $-\sqrt{4} + 3 = 1$. Errors: sign of h; reflection missed; sign of k.

6. $\sqrt[3]{x + 1} = 4$, so $x + 1 = 64$. Errors: cube not undone (3); squared instead of cubed (15); 3 subtracted instead of added (0).

7. Decay factor $1 - 0.06 = 0.94$; 18 months $= 1.5$ years. Errors: months used as years; loss rate used as the factor; growth factor.

8. $g(f(x)) = \frac{5x + 10 - 10}{5} = x$ and $f(g(x)) = x$. Errors: first pair, inverse operation applied to the constant only, so $g(f(x)) = x$ fails (gives $x + 4$); second pair, $+4$ undone by -4 instead of -8 (gives $x + 4$); third pair, sign of $-2x$ lost (gives $-x$).

9. $\frac{\log 32}{\log 8} = \frac{5}{3}$. Errors: ratio inverted; $32 \div 8$; $\log_2 32 - \log_2 8$.

10. A principal square root is never negative. Others: $x = -31$ (a cube root can be negative), $x = 13$, $x = -\frac{1}{2}$.

11. Need $5 - x > 0$; the asymptote is where the argument is 0. Errors: inequality direction; sign of 5; asymptote taken from the $+2$.

12. $f(0) = -\frac{2}{3} + 5 = \frac{13}{3}$; the shift is $k = 5$. Errors: 3^1 used for 3^{-1} (-1); asymptote taken as a; shift in the exponent ignored (3).

13. Asymptote $x = -3$ gives $x + 3$; $\log_2 1 = 0$ at $x = -2$ and $\log_2 4 = 2$ at $x = 1$. Errors: sign of h; shift put outside; base 4 taken from the argument.

14. $y = \sqrt[3]{x} + 4$: undo $+4$ first, then cube. Errors: sign of 4 kept; steps in the wrong order; no inverse step.

15. $(f \circ g)(x) = \sqrt{x^2 - 4}$, so $x^2 \geq 4$. Errors: negative values dropped; inequality reversed; only the domain of f used.

Two-step or DOK 2–3 items: 6, 7, 8, 10, 12, 15.

Sections B and C (0.5 mark per part, 4 marks)

B: a–D $(\log_2 8 = 3)$, b–B $(9^{3/2} = 27)$, c–C $(\log_5 25 = 2)$, d–A (5^{-3}); E is not used. **C:** a. $\log_5 x$. b. 3. c. b. d. $\frac{3}{5}$.

Section D: short answer (4.5 marks)

1. (a) The student multiplied $16 \cdot \frac{3}{4} = 12$; the root must come first: $\left(\sqrt[4]{16}\right)^3 = 8$ [0.5]. (b) $\frac{8x^9}{y^3}$ [0.5].

2. (a) $\sqrt{x + 5} = x - 1$ [0.25]; $x + 5 = x^2 - 2x + 1$ [0.25]; $x^2 - 3x - 4 = 0$, $(x - 4)(x + 1) = 0$, so $x = 4$ or $x = -1$ [0.5].

3. (b) $x = 4$: $\sqrt{9} + 1 = 4$, true [0.5]. $x = -1$: $\sqrt{4} + 1 = 3 \neq -1$, extraneous [0.5]. Only $x = 4$.

4. (a) $(g \circ f)(x) = \left(\sqrt{x + 1}\right)^2 - 1 = x$, domain $x \geq -1$ [0.5]. (b) $(f \circ g)(x) = \sqrt{x^2 - 1 + 1} = \sqrt{x^2} = |x|$ [0.5]; this equals x only when $x \geq 0$, and the range of f is $y \geq 0$, so g must be restricted to $x \geq 0$ [0.5].

Section E: problem solving (4 marks)

(a) $y = 30$ [0.25]: the coffee approaches room temperature, 30°C, and never goes below it [0.25]. **(b)** $T(0) = 30 + A = 94$, so $A = 64$ [0.5]; $T(5) = 30 + 64\left(\frac{1}{2}\right)^{5/h} = 62$ gives $\left(\frac{1}{2}\right)^{5/h} = \frac{1}{2}$, so $h = 5$ [0.5]. **(c)** $T(10) = 30 + 16 = 46$, $T(15) = 30 + 8 = 38$, $T(20) = 30 + 4 = 34$ [1; 0.5 if two correct]. **(d)** Dashed line $T = 30$ [0.25]; points $(0, 94), (5, 62), (10, 46), (15, 38), (20, 34)$ [0.5]; smooth decreasing curve flattening toward the asymptote [0.25]. **(e)** The excess halves each 5 minutes $(64, 32, 16)$, so the drop in the next 5 minutes is 16°C, not 32°C: $T(10) = 46$. The coffee only approaches 30°C [0.5].

Scoring

Total 20 marks. 16 or more: ready. 12 to 15.5: redo the examples for every lost mark, then retry the section. Below 12: re-read the guide lesson by lesson before another attempt.